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TrueRehab-RF

AI-Powered mmWave Radar System for True Recovery Detection and Compensatory Movement Prediction

From Movement Detection to True Recovery Prediction

Prasarnmit Demonstration School (Secondary)
SRINAKHARINWIROT UNIVERSITY

AI Intelligence

77 GHz mmWave Radar

Rehabilitation Support

Contactless Monitoring



1 THE PROBLEM

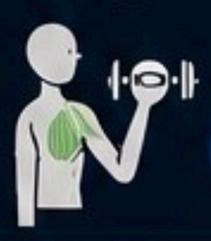
Patients may complete exercises, but they might be compensating with other muscles instead of truly recovering.

- ⚠ Delayed recovery
- ⚠ Re-injury risk
- ⚠ Inefficient rehabilitation

Example: Elbow Flexion Task

TRUE RECOVERY

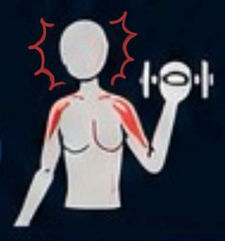
Target muscle works properly. Good control and full range.



VS.

COMPENSATED SUCCESS

Shoulder hike and trunk sway are used to complete the task.



2 OUR INNOVATION

TrueRehab-RF uses a single 77GHz mmWave Radar and AI to monitor micro-movements, analyze movement quality, and predict compensatory patterns in real time.

77 GHz mmWave Radar

Radar Sensing → Signal Processing → AI Analysis → Recovery Assessment

- ✓ Contactless. Intelligent. Personalized.
- ✓ Detects both muscle fatigue and compensatory movement.

3 HOW THE SYSTEM WORKS

A single mmWave Radar captures both micro and macro movements.

MICRO (Muscle Level)
Muscle Vibration & Fatigue

- Muscle vibration
- Frequency analysis
- Fatigue estimation

+

MACRO (Body Level)
Compensatory Movements

- Shoulder elevation
- Trunk sway
- Range of motion
- Movement quality

AI FUSION & EVALUATION

Combine micro and macro features for comprehensive assessment.

Muscle Fatigue | Movement Quality | Compensation Detection | Recovery Score

4 FROM DETECTION TO PREDICTION

Our AI analyzes temporal patterns to predict risk before compenstation becomes obvious.

Muscle Fatigue → Movement Degradation → Compensatory Movement

PREDICTION MODEL (Temporal AI)

Predicts the transition from fatigue to compenstation with an early warning.

EARLY WARNING

Helps therapists adjust the program in time to prevent wrong movement patterns.

5 DEMO SCENARIO: ELBOW FLEXION (BICEPS CURL)

Patient A – True Recovery



Muscle Activity	Range of Motion	Compensation	AI Verdict
78–82%	85–90°	Low	TRUE RECOVERY

⚠ Patient B – Compensated Success



Muscle Activity	Shoulder Hike	Trunk Sway	AI Risk Score
50–55%	+25 mm	15–18°	High

Real-time Dashboard



Muscle Fatigue **High**
Movement Quality **Decreasing**
Compensation Risk **High**
AI Verdict **Compensated Success**

6 PROTOTYPE & VALIDATION PLAN

Data Collection
Collect data from healthy volunteers during rehabilitation exercises.

Ground Truth
sEMG for muscle fatigue
IMU/Video for movement quality

AI Development
Feature extraction, model training, and temporal prediction.

Validation
Compare with ground truth. Evaluate accuracy of detection and prediction.

7 EXPECTED IMPACT

- Improve accuracy of rehabilitation monitoring
- Reduce re-injury risk
- Support personalized rehabilitation
- Enable early intervention
- Advance oral health & rehabilitation innovation

8 KEY FEATURES

- ✓ Contactless monitoring
- ✓ Single 77GHz mmWave Radar
- ✓ Muscle fatigue estimation
- ✓ Compensatory movement detection
- ✓ Early risk prediction
- ✓ Personalized feedback

PROJECT MEMBER

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Let's make rehabilitation smarter.
Let's make recovery truly real.

TRUE RECOVERY,
not just successful compensation.